

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0702

Course Code	CSA0702	Course Title	Computer Networks
CO Assessed	CO3	Assessment	Case Study
Weightage		Total Marks	

Student Name	V.Sree Ruthin Reddy	Reg.No.	192324112
Section/Batch	AIDS-2023	Date of Submission	
Faculty Incharge	DR.R S Selvan	Assessment	Individual Submission

TOTAL MARKS	20	OBTAINED MARKS	20
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COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I V.Sree Ruthin Reddy(192324112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sree Ruthin Reddy

Course Faculty
Dr.R.Senthamil Selvan

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching, and TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values, Anycast routing, and DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control, Nagle's algorithm, and SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 - cloud Based Video Conferencing.

1. The performance issue is caused by both the transport protocol (TCP/UDP) or the application layer. but the primary cause is the network conditions affecting UDP communication.

Video conferencing applications use UDP because it provides low latency by transmitting packets without acknowledgements or retransmissions. However, UDP cannot recover lost packets. The reported 8% packet loss

2. Real-time multi-media applications have strict Quality of Service (QoS) requirements:

Low Latency: Voice and video should reach users almost instantly.

Low jitter: packets must arrive at regular intervals.

Minimal packet loss: lost packets reduce audio and video Quality.

Continuous communication: Delayed retransmissions are less acceptable than occasional packet loss.

3. Protocol-level Improvements

- Continue using UDP with RTP for efficient media delivery.
- Implement Forward Error Correction
- Configure Quality of service to prioritize video conferencing traffic

2.

CASE STUDY - 2

The observed DNS lookup time of 800 ms is significantly higher than acceptable values.

- Centralized DNS server located far from users.
- Multiple recursive DNS lookups before reaching the authoritative server.
- Poor DNS caching resulting in repeated queries.

② A distributed DNS architecture should include:

- Deploy multiple DNS servers globally using the same IP addresses.
- users are automatically directed to the nearest DNS server.

③ TTL advantages & limitations.

- Faster DNS resolution
- Improved scalability for global users.
- Automatic failover during server failures.
- Reduced latency.
- Lower workload on authoritative DNS servers.

The proposed architecture provides excellent performance, scalability and availability while significantly improving user experience.

CASE STUDY - 3

4. A centralised DNS architecture directs all DNS requests to a single primary server.

Impact on performance:

- Creates a single point of failure
 - High traffic causes server overload.
 - DNS lookup increases to 800ms.
 - Delays application access.
 - Increases response time reduces customer satisfaction.
5. • Multiple authoritative DNS servers across continents:
- Anycast routing to connect users to the nearest server.
 - Recursive DNS caching at regional locations
 - Optimized TTL values to reduce

6. TTL (Time-to-Live)

- Controls cache duration
- Reduces repeated DNS queries.
- Improves response time.
- Excessively long TTL delays record updates.

Combining optimized TTL, Anycast routing and DNS caching provides low latency, high availability, fault tolerance.

4

Case study-4

TCP flow control prevents the sender from overwhelming the receiver. During heavy trading, reduced receiver window size forces the sender to wait before transmitting additional data, increasing latency.

→ Nagle's Algorithm

→ SYN Queue Limitations

2. Flow control

- slows data transmission
- delays order acknowledgements
- Reduces transaction throughput.

These issues increase order acknowledgement latency from ms to to ms, negatively affecting trading accuracy and customer satisfaction.

3. Recommended improvements

- Disable Nagle's Algorithm using `TCP_NODELAY`
- Increase TCP SYN backlog
- Enable TCP fast open
- Increase TCP send and receive buffer sizes.